(the surface energy). In the fifth term, which accounts for the odd-even effect, a positive sign is used for even proton-even neutron nuclei and a negative sign for odd proton-odd neutron nuclei. For nuclei of odd A (even-odd or odd-even) this term has the value of zero. Comparison of this equation with actual binding energies of nuclei yields a set of coefficients; e.g.

$$a_v = 15.5, \quad a_a = 23, \quad a_c = 0.72, \quad a_s = 16.8, \quad a_\delta = 34$$

With these coefficients the binding energy equations (4.5) and (4.10) give agreement within a few percent of the measured values for most nuclei of mass number greater than 40.

When the calculated binding energy is compared with the experimental binding energy, it is seen that for certain values of neutron and proton numbers, the disagreement is more serious. These numbers are related to the so-called "magic numbers", which we have indicated in Figure 4.1, whose recognition led to the development of the nuclear shell model described in a later chapter.

4.7. VALLEY OF β-STABILITY

If the semiempirical mass equation is written as a function of Z, remembering that $N = A - Z$, it reduces to a quadratic equation of the form

$$E_B = a\,Z^2 + b\,Z + c \pm d/A^{3/4} \tag{4.16}$$

where the terms a, b and c also contain A. This quadratic equation describes a parabola for constant values of A. Consequently, we would expect that for any family of isobars (i.e. constant A) the masses should fall upon a parabolic curve. Such a curve is shown in Figure 4.5. In returning to Figure 4.1, the isobar line with constant A but varying Z cuts

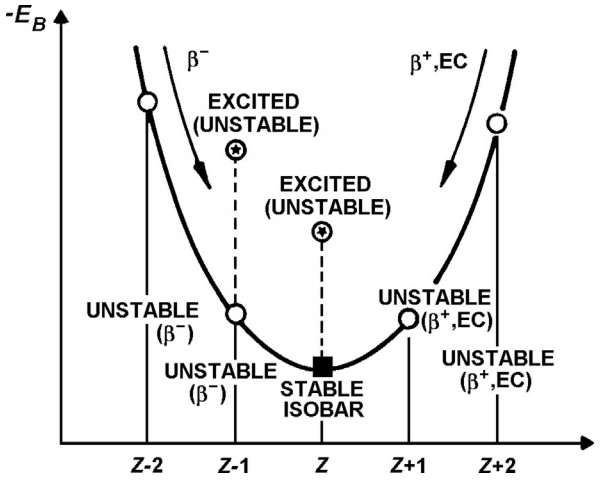


Figure 4.5 Isobar cut across the valley of stability showing schematically the position of different kinds of nuclei.

diagonally through the line of stable nuclei. We can picture this as a valley, where the most stable nuclei lie at the bottom of it (cf. Figs. 4.1 and 4.5), while unstable nuclei lie up the valley sides as shown in Figure 4.5. Any particular isobaric parabola can be considered as a cross-section of the valley of stability; Figure 4.5 would be seen by someone standing up to the right of Figure 4.1 and looking down the valley. The isobars located on the sides of the parabola (or slope of the valley) are unstable to radioactive decay to more stable nuclides lower on the parabola, though usually the most stable nucleus is not located exactly at the minimum of the parabola. Nuclides on the left hand side of the parabola (lower atomic numbers) are unstable to decay by β^-emission. Isobars to the right of the valley of stability are unstable to β^+ decay or electron capture. At the bottom of the valley the isobars are stable against β decay. The curved line in Figure 4.1 is calculated for maximum stability according to (4.15), and indicates the theoretical bottom of the valley. The minimum of the curve can be calculated from (4.15) to be

$$Z = 2A/[4 + (a_c/a_a)A^{2/3}] \tag{4.17}$$

and is shown in Figure 4.1. For small A values (4.17) reduces to $Z = A/2$ or $N = Z$; thus the bottom of the stability valley follows the $N = Z$ line as indicated in Figure 4.1 for the lighter nuclides.

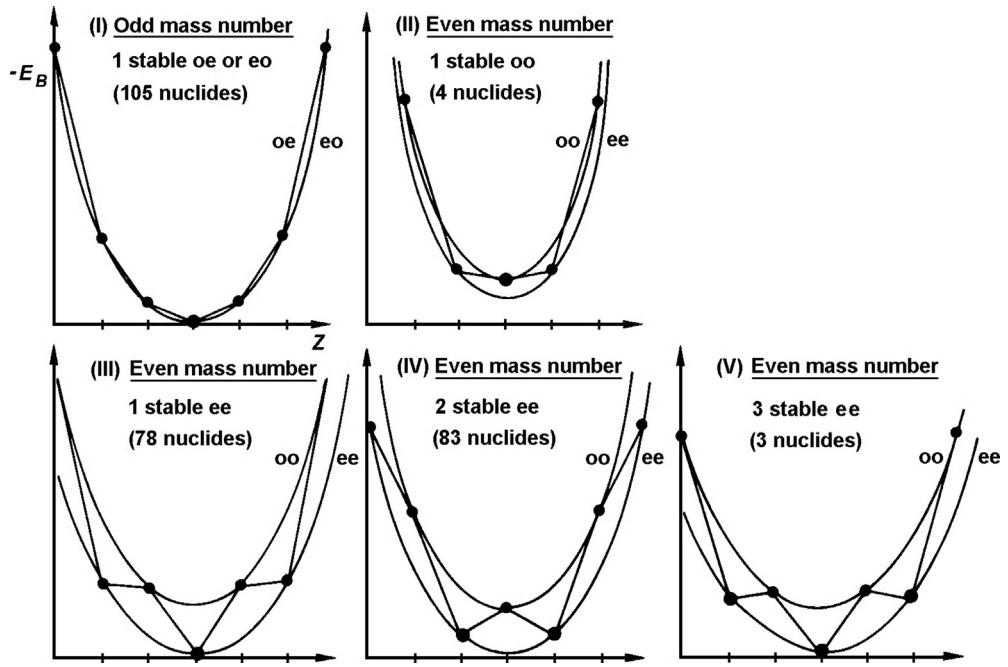


Figure 4.6 Isobar parabolas for odd mass numbers (I: odd-even or even-odd nuclides) and for even mass numbers (cases II - V). The stable nuclides are indicated by heavier dots.

A closer analysis of (4.16) makes us expect that the last term gives rise to three different isobaric parabola depending on whether the nuclei are odd-A (even-odd or odd-even), odd-odd, or even-even (Fig. 4.6). In the first case, in which the mass number is odd, we find a single parabola (I); whether all beta decay leads to changes from odd-even to even-odd, etc. For even mass numbers one finds a double parabola (II) – (V). When the individual nuclear properties are considered, the difference between the curves for the odd-odd and even-even nuclei may lead to alternatives with regard to the numbers of possible stable isobars: it is possible to find three stable isobars (case V) although two (case IV) are more common. Although the odd-odd curve always must lie above the even-even curve, still an odd-odd nucleus may become stable, as is shown for case II.

4.8. THE MISSING ELEMENTS: $_{43}$TC AND $_{61}$PM

Among the stable elements between $_1$H and $_{82}$Pb two elements are "missing": atomic number 43, named *technetium* (Tc), and atomic number 61, *promethium* (Pm). Though these elements can be produced through nuclear reactions and also have been found to exist in certain stars, they are not found on earth because their longest lived isotopes have much too short half-lives for them to have survived since the formation of our planet. This can be understood by considering the valley of β-stability. For pedagogic reasons we will first discuss promethium.

4.8.1. Promethium

The valley of β-stability for $Z = 61$ shows a minimum around mass number $A = 146$, for which the isotopes are either of the even-even or of the odd-odd type. Thus the binding energy curve should exhibit two isobar parabolas, as illustrated in Figure 4.7; the decay energy Q is released binding energy. ^{146}Pm has a 5.5 y half-life and decays either by electron capture (63%) to ^{148}Nd or by β^--emission (37%) to ^{146}Sm, who both are more stable (i.e. have a larger nucleon binding energy); the nuclear binding energy is given on the vertical scale in the Figure. The curves shown in Figure 4.7 differ from those calculated from eqn. 4.14 by about 1 unit in Z due to deviations from the simple liquid drop model in the lanthanide region, see §6.4.

The two adjacent isobars, mass numbers $A = 145$ and $A = 147$, are of the odd-even and even-odd types, thus only one isobaric β-decay curve exists for each of these. The decay scheme for $A = 145$ follows curve I in Figure 4.6 for which $^{145}_{60}$Nd is the stable isobar. ^{145}Pm is the longest lived promethium isotope ($t_{½}$ 17.7 y). For $A = 147$, the stable isobar is $^{147}_{62}$Sm; the half-life of ^{147}Pm is 2.62 y, which makes it the most convenient radioisotope of promethium for use in experiments.

Promethium is a fission product (Ch. 5 and 19) and can be chemically isolated in pure form. It exhibits typical lanthanide properties and is used in technology and medicine as a radiation source (Ch. 18).